

Distance sensors (and reference frames)

Sensors

Physical Property	→	Sensing Technology
Contact	→	bump, switch
Distance	→	ultrasound, radar, infra red
Light level	→	photocells, cameras
Sound level	→	microphones
Strain	→	strain gauges
Rotation	→	encoders and potentiometers
Acceleration	→	accelerometers and gyroscopes
Magnetism	→	compasses
Smell	→	chemical sensors
Temperature	→	thermal, infra red
Inclination	→	inclinometers, gyroscopes
Pressure	→	pressure gauges
Altitude	→	altimeters

When would we use a distance sensor?

- 1. Obstacle detection and avoidance**
- 2. Position control**
- 3. Mapping and navigation**
- 4. Human or object detection**
- 5. Measurement and quality control**

Passive vs active sensors

Passive Sensors

- Only receive information; they do **not emit** any signal.
- Example: **Bump sensors**, temperature sensors, light sensors.
- **Advantages:**
 - Lower cost
 - Lower power consumption

Active Sensors

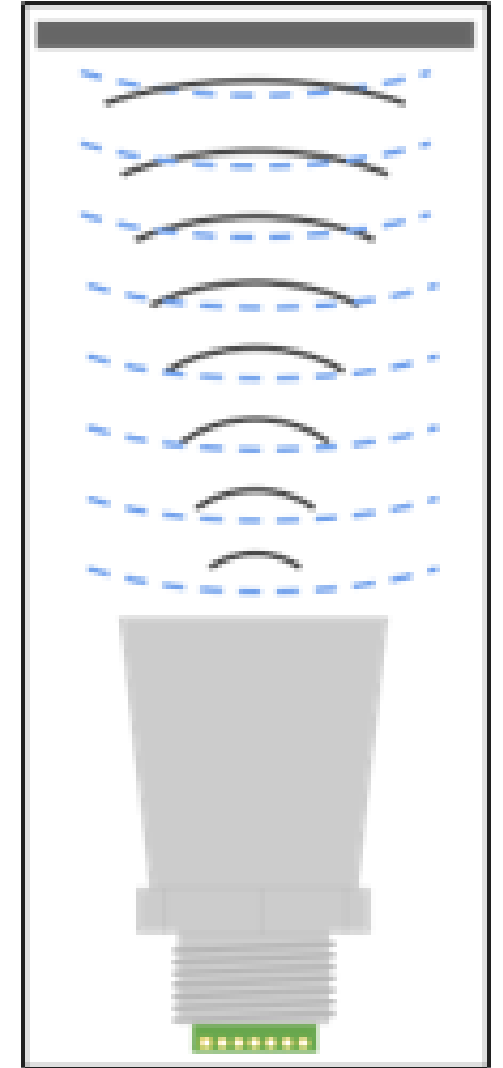
- **Send out** a signal (such as sound, light, or radio waves) and **measure** the reflection or response.
- Example: **Ultrasonic sensors**, **Radar**, **LiDAR**.
- **Advantages / Disadvantages:**
 - Higher accuracy and can work in dark or complex environments
 - More expensive
 - Higher power consumption

Ultrasonic sensors

- Has a “speaker” (emitter) and a “microphone” (detector)
- Sends out a sound wave that bounces off an object and returns
- Uses the time to return and the speed of sound to calculate distance



HC-SR04, \$1.50



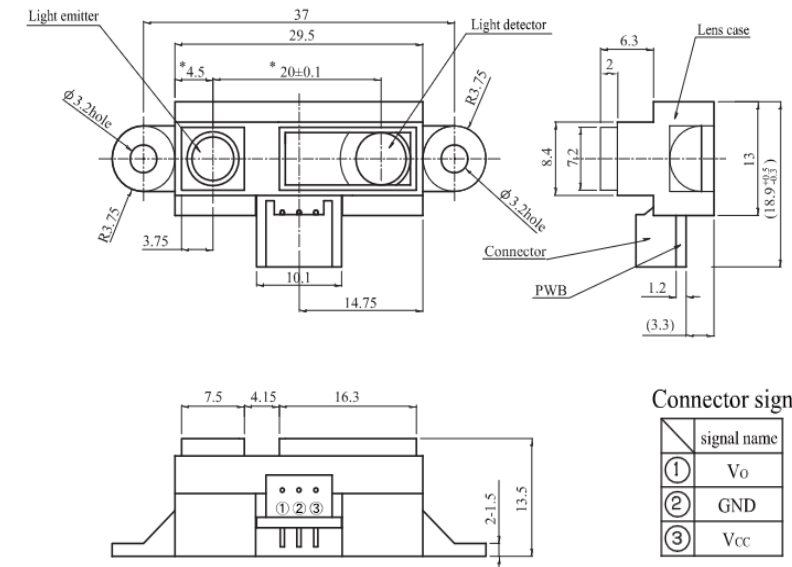
$$d = \frac{\text{speed of sound}}{2} \times \Delta t$$

Infrared (IR) sensors

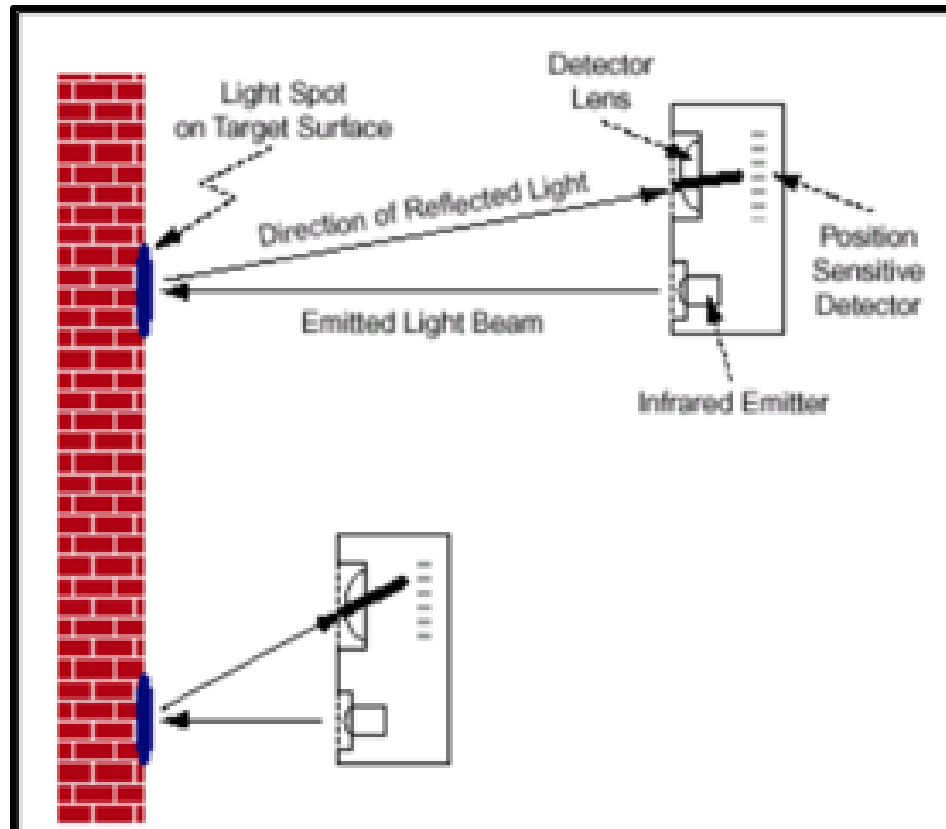
- Uses IR emitter and detector
- Detector is a Position Sensitive Detector (PSD)
- Angle of return radiation determines position on detector
- This is used to calculate the distance
- Farther distance = smaller angle to detector



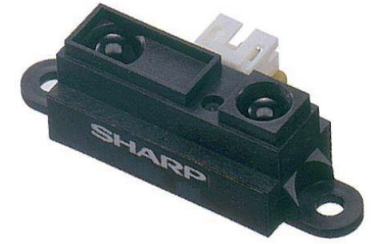
SHARP IR sensor, \$15



Infrared (IR) sensors



IR sensor calibration

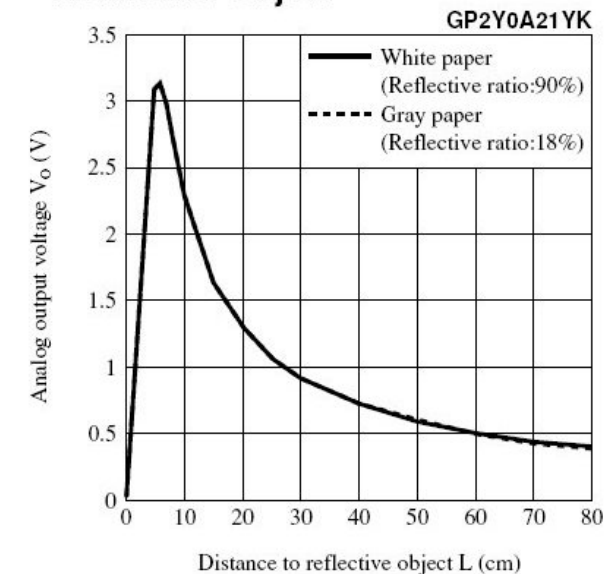


- Calibration is used to find the **mathematical relationship** between the **sensor's analog output voltage (or ADC reading)** and the **actual distance to an object**.
- Take measurements over the range of operation
- Arduino analog pins, A0-A5
- Find a best fit line to the data
- See Datasheet
 - Inverse distance relationship
 - Relationship between distance and voltage can be approximated as:

$$\frac{1}{d+k} = m \times ADC + b$$

d = distance,
 ADC = analog reading,
 m, b, k = calibration constants.

Fig.5 Analog Output Voltage vs. Distance to Reflective Object



$$d = \frac{c \times \Delta t}{2}$$

Time of Flight (ToF) sensors

- Multiple methods

- A Time of Flight (ToF) sensor measures distance by calculating the time it takes for a light pulse (usually infrared or laser) to travel to an object and back.
- Amplitude modulation, check phase shift
- Pulse square wave, check phase shift

where

- d = distance
- c = speed of light
- Δt = time between emission and detection

Working Methods:

1. Direct Time Measurement:

- Measure the **exact travel time** between emission and reflection.
- Very accurate but requires **fast and expensive electronics**.

2. Amplitude Modulation (Phase Shift):

- Modulate the emitted light wave and **compare phase shift** between sent and received signals.
- Works well but **slower** than direct methods.

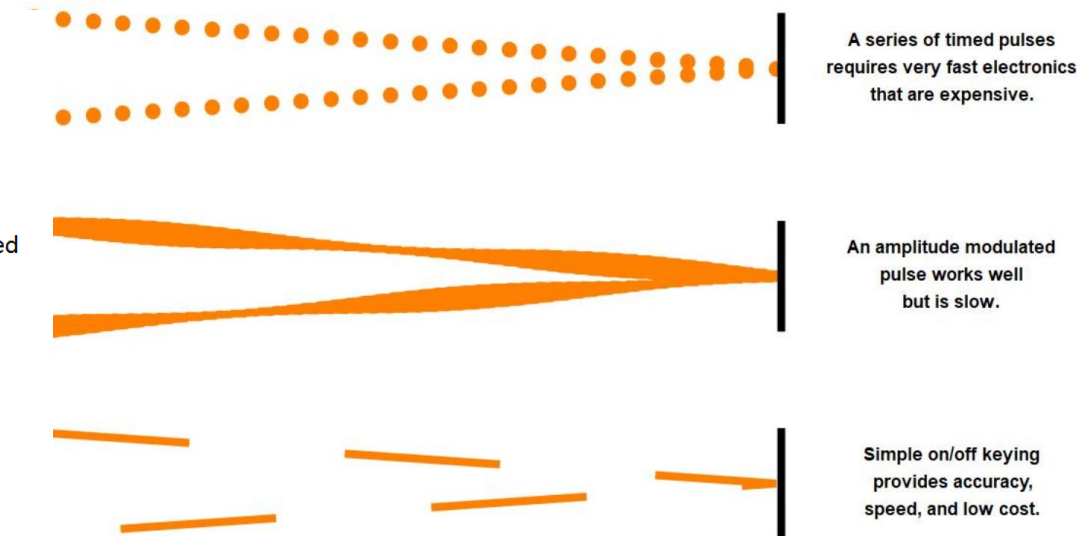
3. Square Wave Modulation (Simplified Phase Shift):

- Use a square wave signal and check the **phase difference**.
- Provides a **good balance** of accuracy, speed, and cost.



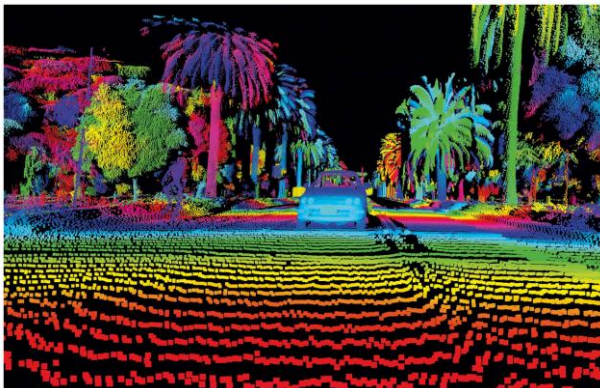
TF Mini Plus, \$50

Difference are amplitude and frequency

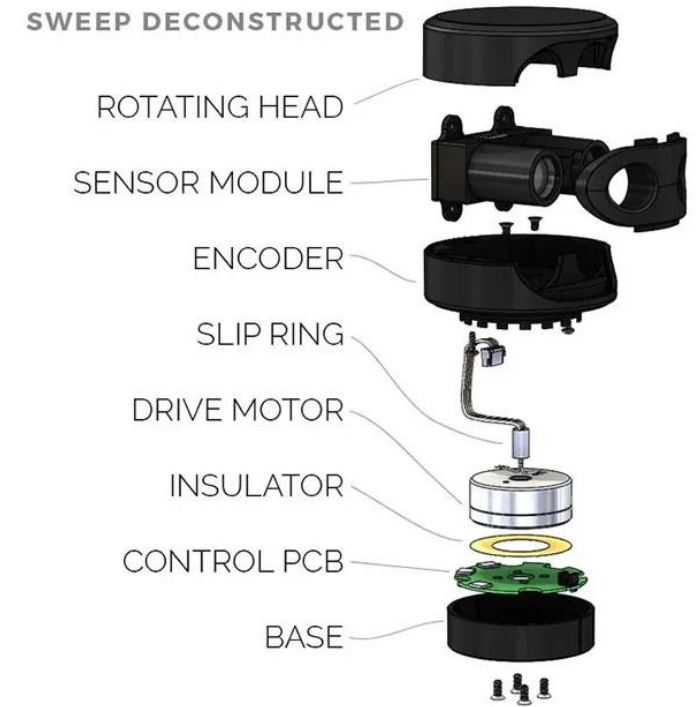


LiDAR

- LiDAR is a **remote sensing technology** that measures **distance** by sending out laser pulses and measuring the **time of flight** (ToF) of the reflected light.
- Light detection and ranging
- Rotating laser time-of-flight sensor(s)
- Generates point cloud data with high frequency



<https://www.youtube.com/watch?v=3EehCU3csJQ>



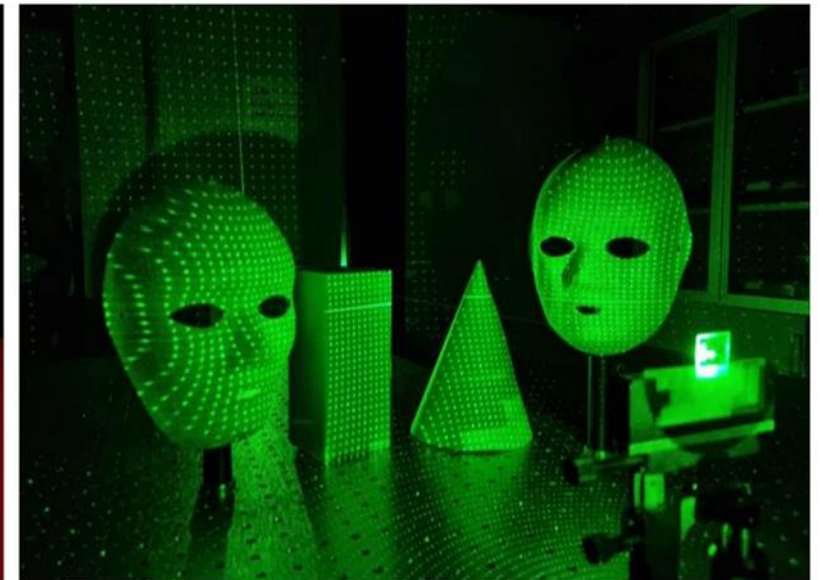
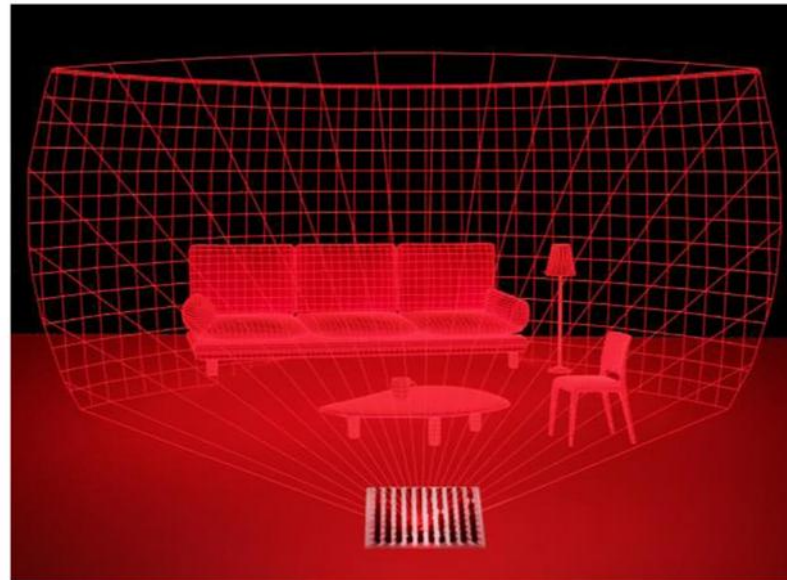
Ouster LIDAR \$3,500-20,000

3D cameras

- Structured light arrays or **time of flight array**
- Pattern of lights projected onto subject
- Distortion of the pattern is used to measure shape



Orbecc Astra \$150



Compare and contrast

- Infrared
 - Better at detecting edges
 - Not good for outdoors (sunlight interferes)
 - Analog output, affected by supply voltage
- Ultrasonic
 - Robust against varying light and dust conditions
 - Detect blobs, not good with edges
 - Time-based signal, not strongly influenced by supply voltage
- Time of Flight
 - Quick response
 - Can be placed in arrays for 3D
 - Complex hardware, large power requirements

Factors to consider in sensor selection

- Power requirements
- Sensor accuracy
- Resolution requirements
 - Spatial resolution
 - Measurement resolution
 - Time resolution
 - Frequency response
- Cost

Local measurements in a global reference frame

Next time

Transformation Matrices

Rotation Matrices